KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade
Anand Nagar, Krishnankoil, Srivilliputtur, Virudhunagar (Dt.) - 626126 | OMR, Chennai, Tamil Nadu
info@kalasalingam.ac.in | www.kalasalingam.ac.in

# InterviewGen AI -
# AI-Based Mock Interview and Performance Analyzer

**Project Documentation**

**Project ID:** PRJ086

**Team ID:** T086

**Team Name :** Byte-Bloom

**Team GitHub Repository:** https://github.com/ChSivamowny15/InterviewGen-AI

**Live Link :** https://interviewgen-ai-rtnk.onrender.com



| Student Name | Registration Number | Individual GitHub Link |
|---|---|---|
| KANIKARAM MEDHARA POOJITHA | 99230040116 | github.com/KANIKARAMPOOJITHA |
| CHEEMALA SIVA MOWNY | 99230040092 | github.com/ChSivamowny15 |
| GAJULA HEMALATHA | 99230040300 | github.com/99230040300 |
| GANDHAM PRAVALLIKA | 99230040302 | github.com/pravalikagandham4 |



**Project Trainer / Guide**

**Mr. SHIRWIN PRINCE I**

## Table of Contents

1.Introduction……...Section1
2. Problem Statement, Objectives & Scope…… ……....Section2
3. System Architecture and RAG Pipeline .....Section 3
4. Implementation Details and Technologies Used ….....Section 4
5. Results, Testing and Performance Evaluation .....Section 5
6.Main Features……Section 6
7. Conclusion and Future Scope .....Section 7
8. References .....Section 8

## 1.Introduction

InterviewGen AI – AI-Based Mock Interview and Performance Analyzer is an intelligent web-based application designed to help students and job seekers improve their interview preparation using Artificial Intelligence. In today's competitive job market, candidates require strong technical knowledge, communication skills, and confidence to succeed in recruitment interviews. However, limited practice opportunities and the lack of personalized feedback make interview preparation challenging. To address these issues, InterviewGen AI provides an interactive platform for conducting mock interviews based on different job roles, interview categories, and difficulty levels. It evaluates candidate responses and provides personalized feedback to identify strengths and areas for improvement.

The system integrates **Retrieval-Augmented Generation (RAG)** to retrieve relevant candidate information and provide context-aware assistance, along with **Large Language Model (LLM)** technology through the Groq API using the GPT-OSS-120B model for intelligent response evaluation and feedback generation. It also includes features such as resume analysis, voice-based interview practice, speech recognition, webcam-based monitoring, performance analytics, and interview report generation. Developed using Python, Flask, SQLite, HTML, CSS, and JavaScript, InterviewGen AI aims to provide a user-friendly and effective environment for improving interview skills, building confidence, and preparing candidates for real-world recruitment processes.

## 2. Problem Statement, Objectives & Scope

### 2.1 Problem Statement

Many students and job seekers face difficulties in preparing for technical and HR interviews due to limited practice opportunities, lack of personalized guidance, and insufficient feedback on their performance. Traditional interview preparation methods do not always provide real-time evaluation or identify individual weaknesses in technical knowledge, communication, and confidence. Therefore, there is a need for an intelligent and interactive platform that can simulate interview environments, analyze candidate responses, and provide personalized suggestions for improvement using Artificial Intelligence.

### 2.2 Objectives

The main objectives of InterviewGen AI are:

1. To develop an AI-powered web application that provides interactive mock interviews based on different job roles, interview categories, and difficulty levels.
2. To implement Retrieval-Augmented Generation (RAG) for retrieving relevant candidate information from uploaded resumes and providing personalized, context-aware interview assistance.
3. To extract and process resume content to identify candidate skills, educational qualifications, projects, and technical experience for relevant information retrieval.
4. To integrate the retrieved context with the Large Language Model (LLM) through the Groq API using GPT-OSS-120B to support intelligent interview question generation, answer evaluation, and feedback generation.
5. To provide additional features such as resume analysis, voice-based interview practice, speech recognition, and performance analytics.
6. To generate detailed interview performance reports highlighting candidates' strengths and weaknesses, helping them improve their technical knowledge, communication skills, and confidence.

### 2.3 Scope of the Project

InterviewGen AI focuses on providing a comprehensive interview preparation environment for students and job seekers. The system supports multiple interview categories, including Technical, HR, Behavioral, and Mixed interviews, with different difficulty levels. It offers resume analysis, AI-based answer evaluation, RAG-based context retrieval, voice interaction, webcam-based monitoring, performance analytics, and report generation. The platform is designed to provide accessible and personalized interview practice through a web-based interface. In the future, it can be enhanced with multilingual interview support, advanced speech analysis, improved retrieval mechanisms, and additional job-role-specific interview scenarios.

## 3. System Architecture and RAG Pipeline

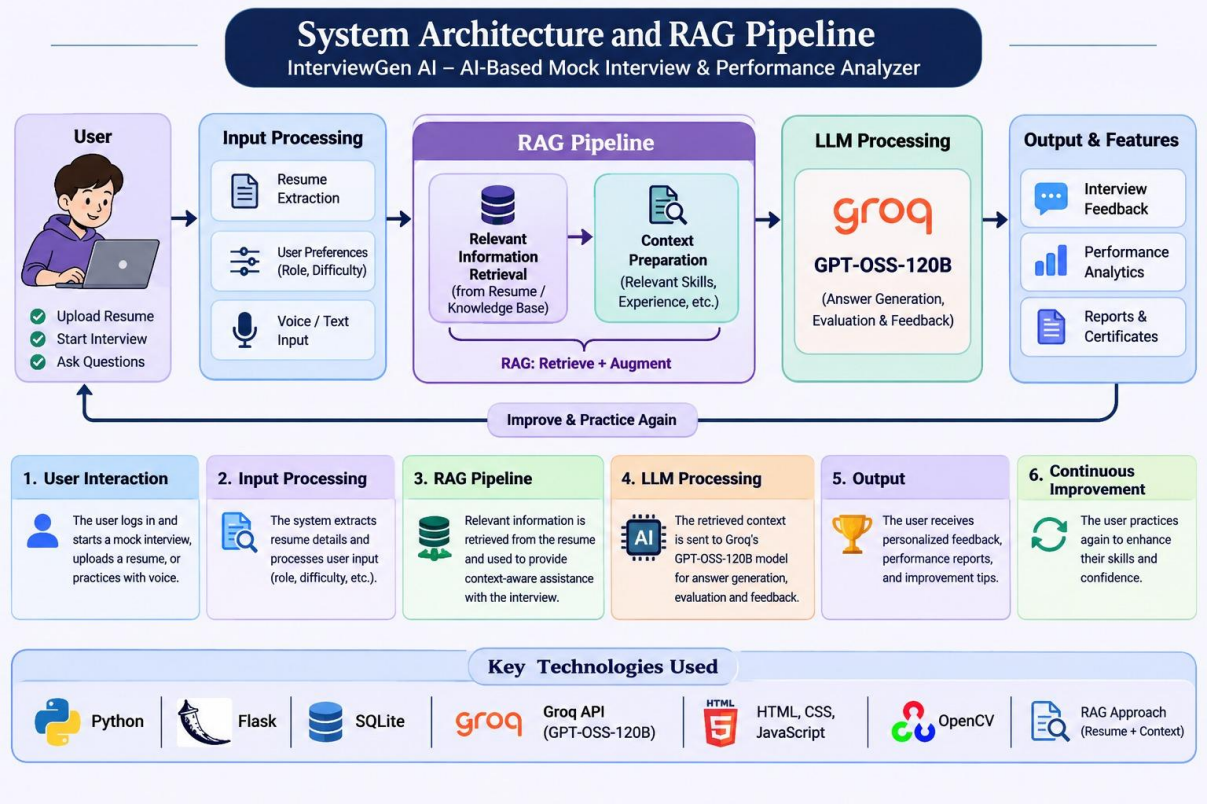



Figure 1: Overall System Architecture and RAG Pipeline of InterviewGen AI

The system architecture of InterviewGen AI consists of several interconnected modules that work together to provide an intelligent interview preparation platform. The user interacts with the application through a web-based interface developed using Flask, where resumes can be uploaded and mock interviews can be started based on selected job roles and difficulty levels. The system processes the candidate's information and

uses the Retrieval-Augmented Generation (RAG) approach to identify relevant information from the resume and prepare contextual inputs for the Large Language Model (LLM).

The retrieved context is provided to the GPT-OSS-120B model through the Groq API to support interview question generation, answer evaluation, and personalized feedback. The application also provides additional functionalities such as resume analysis, voice-based practice, performance analytics, and interview report generation. The overall architecture enables candidates to practice interviews, understand their performance, identify areas for improvement, and enhance their confidence for real-world recruitment processes.

## 4. Implementation Details and Technologies Used

### 4.1 Technology Stack

The following technologies are used to develop InterviewGen AI:

| Component | Technology Used | Purpose |
|---|---|---|
| Frontend | HTML, CSS, JavaScript | User interface, styling, and interactive features |
| Backend | Python, Flask | Application logic, routing, and server-side processing |
| Database | SQLite | Storage of user information and interview records |
| AI Model | GPT-OSS-120B | AI-based question generation, answer evaluation, and feedback |
| AI Integration | Groq API | Connecting the application with the LLM |
| RAG | Resume-based context retrieval | Retrieving relevant candidate information for |

| | | |
|---|---|---|
| | | personalized interview assistance |
| Resume Processing | PyPDF, python-docx | Extracting text from PDF and DOCX resumes |
| Image Processing | OpenCV | Webcam-based image processing |
| Voice Processing | Speech Recognition, Text-to-Speech | Voice-based interview interaction |
| Development Environment | Visual Studio Code | Application development and debugging |
| Deployment | Render | Hosting the web application |

### 4.2 RAG Implementation

InterviewGen AI incorporates a Retrieval-Augmented Generation (RAG) approach to provide personalized interview assistance using candidate resume information.

The RAG workflow consists of the following stages:

1. **Document Input:** The candidate uploads a resume in PDF or DOCX format.
2. **Text Extraction:** Resume content is extracted and processed to identify relevant information.
3. **Context Retrieval:** Relevant candidate information is retrieved according to the interview context.
4. **Context Augmentation:** The retrieved information is included in the prompt provided to the language model.

5. **Response Generation:** The GPT-OSS-120B model processes the contextual input through the Groq API and generates relevant interview assistance and feedback.

### 4.3 Database Design

SQLite is used as the database management system to store and manage application information.

| Data Entity | Information Stored | Purpose |
|---|---|---|
| User Information | User ID, name, email, and account details | User registration and authentication |
| Interview Records | Interview details, selected role, category, and difficulty | Maintaining interview history |
| Interview Evaluation | Candidate answers, scores, and feedback | Performance evaluation |
| Resume Information | Extracted resume-related information | Supporting resume analysis and personalized assistance |

### 4.4 AI Model Integration

InterviewGen AI integrates Generative Artificial Intelligence through the Groq API, which provides access to the GPT-OSS-120B Large Language Model (LLM). The model is used to perform intelligent text generation and language-based analysis within the application. It plays an important role in generating interview questions, evaluating candidate responses, and providing personalized feedback based on the information supplied by the application.

When a candidate starts a mock interview, the application collects the selected job role, interview category, and difficulty level. These details are used to prepare the input prompt for the AI model, enabling it to generate relevant interview questions. During

the interview, the candidate submits answers, which are sent to the model for analysis. The model evaluates the responses based on their relevance, clarity, correctness, and overall quality, and generates feedback with suggestions for improvement.

The application also incorporates a Retrieval-Augmented Generation (RAG) approach to support context-aware responses. Relevant information extracted and retrieved from the candidate's uploaded resume can be included in the prompt provided to the LLM. This helps the model generate interview questions and feedback that are more closely related to the candidate's skills, educational qualifications, and project experience. By combining retrieved information with the language generation capabilities of the model, the application provides more personalized interview assistance.

The Groq API acts as the communication interface between the Flask backend and the GPT-OSS-120B model. The backend prepares the required prompts, sends requests to the API, receives the generated responses, and processes the results before presenting them through the web interface. This integration enables the application to deliver AI-powered interview assistance without requiring the language model to run locally.

Overall, the integration of the Groq API, GPT-OSS-120B model, and RAG approach forms the core intelligence of InterviewGen AI. It enables automated question generation, response analysis, and personalized feedback, making the interview preparation process more interactive and context-aware.

### 4.5 Technology Selection

The technologies used in InterviewGen AI are selected based on their simplicity, flexibility, reliability, and suitability for developing an intelligent web-based application. Python is used as the primary programming language because of its easy-to-understand syntax and extensive support for web development and Artificial Intelligence. Flask is selected as the backend framework to handle application requests, manage server-side operations, process user inputs, and connect different modules. HTML, CSS, and JavaScript are used to develop an interactive and user-friendly frontend interface.

SQLite is used for database management because it is lightweight, easy to configure, and suitable for storing user information, interview records, candidate responses, and evaluation results. The Groq API is integrated to access the GPT-OSS-120B Large Language Model, which supports Generative AI functionalities such as interview

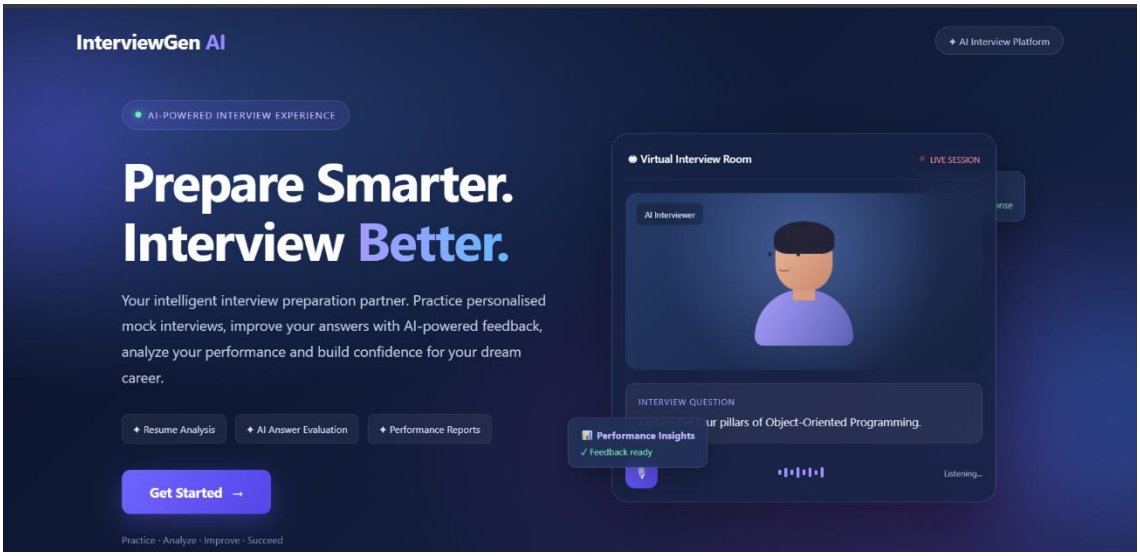
question generation, candidate answer evaluation, and personalized feedback. Supporting libraries are also used for resume document processing, voice interaction, and webcam-based functionalities.

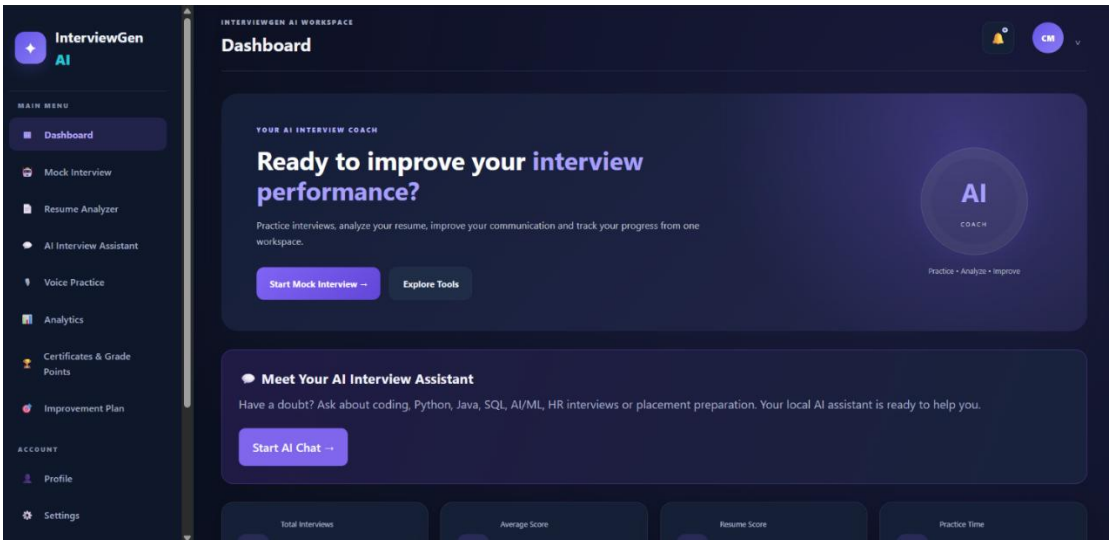
The Retrieval-Augmented Generation (RAG) approach is incorporated to retrieve relevant information from candidate resumes and provide it as contextual input to the language model. This helps generate interview-related questions and responses that are more relevant to the candidate's skills, educational background, and project experience. The combination of these technologies provides an integrated technical foundation for InterviewGen AI, supporting intelligent interview practice, personalized assistance, and performance analysis through a single web-based platform.

## 5.Overview of the System or System architecture

### 5.1 Results

InterviewGen AI was successfully developed as an AI-powered web application for interview preparation and performance analysis. The application provides an interactive platform where users can register, log in, upload resumes, and participate in mock interviews based on different job roles and difficulty levels. The RAG approach supports resume-based context retrieval, while the GPT-OSS-120B model through the Groq API provides AI-based question generation, answer evaluation, and personalized feedback. Additional features include voice-based practice, resume analysis, webcam-based monitoring, performance analytics, and report generation.






## 5.2 Testing

| Test Module | Testing Performed | Expected Result |
|---|---|---|
| User Authentication | Registration and login | Successful account access |
| Resume Analysis | Uploading PDF and DOCX resumes | Resume information extracted |
| RAG Pipeline | Retrieving relevant resume information | Relevant candidate context provided to the LLM |
| AI Interview | Generating interview questions | Relevant questions generated |
| Answer Evaluation | Submitting candidate responses | Scores and feedback generated |

### 5.3 Performance Evaluation

The performance of InterviewGen AI is evaluated based on the functionality, responsiveness, accuracy, and output quality of its different modules. Each module is examined to understand how effectively it supports the overall interview preparation process. The system provides AI-generated feedback based on candidate responses and uses retrieved resume information to support personalized interview assistance.

The resume analysis module processes the uploaded candidate documents and identifies important details such as technical skills, educational qualifications, projects, and other relevant information. These details help the system understand the candidate's background and provide more relevant interview preparation support. The RAG approach helps retrieve relevant resume information and include it as contextual input for the Large Language Model, improving the relevance of generated responses.

The Generative AI module, integrated through the Groq API using the GPT-OSS-120B model, supports interview question generation, answer evaluation, and personalized feedback. It analyzes candidate responses and provides suggestions to improve answer quality, technical understanding, and communication. This allows candidates to identify mistakes and improve their responses through repeated practice.

The performance analytics module presents candidate results in an organized manner by highlighting strengths, weaknesses, and areas requiring further improvement. The voice-based functionalities support spoken interview practice, while webcam-based functionalities provide additional support during the interview session. The report generation module helps candidates review their interview performance and understand their progress.

Overall, the evaluation demonstrates that the application combines multiple AI-powered functionalities into a single platform. The integration of RAG, Generative AI, resume analysis, voice interaction, and performance analytics provides a structured

environment for interview preparation. The system is designed to offer a user-friendly experience and support continuous improvement in candidates' interview skills.

### 5.4 Overall Outcome

The developed InterviewGen AI application successfully demonstrates the integration of modern web technologies, Retrieval-Augmented Generation (RAG), and Large Language Model (LLM) capabilities to create an intelligent interview preparation platform. The application provides an interactive environment where candidates can prepare for different types of interviews based on their preferred job roles and difficulty levels.

The system uses Generative AI to support interview question generation, candidate response evaluation, and personalized feedback. Through the RAG approach, relevant information from the candidate's resume can be used as context to provide more personalized interview assistance. This helps candidates practice questions related to their skills, educational background, and project experience.

Additionally, the application combines features such as resume analysis, voice-based interview practice, performance tracking, and report generation to support different aspects of interview preparation. The feedback provided by the system helps candidates understand their strengths, identify areas that require improvement, and work towards developing better technical knowledge and communication skills.

Overall, InterviewGen AI provides a structured and user-friendly platform for continuous interview practice. The project demonstrates the practical application of Generative AI and RAG in an educational and career development environment, helping students improve their preparation and build confidence before attending real-world recruitment interviews.

## 6.Main Features

InterviewGen AI offers several intelligent features to support effective interview preparation and performance improvement. The major features of the application include:

1. **User Authentication and Profile Management:** Provides user registration, login, profile management, and secure access to the application.
2. **AI-Based Resume Analysis:** Extracts and analyzes relevant information from uploaded resumes, including technical skills, educational qualifications, and project details.
3. **AI-Powered Mock Interviews:** Conducts interactive mock interviews based on different job roles, interview categories, and difficulty levels.
4. **RAG-Based Context Retrieval:** Retrieves relevant candidate information from uploaded resumes and provides contextual information to the language model for personalized interview assistance.
5. **AI-Based Answer Evaluation:** Uses the GPT-OSS-120B model through the Groq API to evaluate candidate responses and generate scores, feedback, and improvement suggestions.
6. **Voice-Based Interview Practice:** Enables candidates to answer interview questions through voice input using speech recognition and text-to-speech functionalities.
7. **Webcam-Based Monitoring:** Uses OpenCV-based image processing to support webcam-related interview monitoring.
8. **Performance Analytics:** Analyzes interview performance, tracks scores, and identifies candidates' strengths and weaknesses.
9. **Interview Report Generation:** Provides interview performance reports with evaluation results and suggestions for improvement.
10. **AI Chatbot Assistance:** Offers an interactive AI assistant to support candidates with interview preparation and related queries.

11. **Certificates and Grade Points:** The dashboard includes an achievements section where users can view certificates earned and accumulated grade points. These values are expected to depend on completed interview sessions and the project's achievement rules.

## 7. Conclusion and Future Scope

`

### 7.1 Conclusion

InterviewGen AI demonstrates how Artificial Intelligence can be applied to make interview preparation more effective, accessible, and personalized. The project brings together resume-based information retrieval, language model capabilities, and interactive web technologies to create an environment where candidates can practice, evaluate their responses, and learn from their mistakes. By providing structured feedback and performance insights, the system encourages continuous improvement and independent learning. It also demonstrates the practical integration of Retrieval-Augmented Generation (RAG) and Large Language Model (LLM) technologies in an educational and career-development application.

The integration of Generative AI enables the system to provide dynamic interview questions, evaluate candidate responses, and offer suggestions based on individual performance. The use of resume-related context further supports personalized interview preparation by connecting questions and feedback with the candidate's skills, academic background, and project experience. Additional functionalities such as resume analysis, voice-based practice, and performance reporting contribute to a more comprehensive interview preparation experience.

The project also provides practical exposure to the development of an AI-powered web application by combining backend programming, frontend technologies, database management, and external AI model integration. It highlights the importance of

combining information retrieval with language generation to develop context-aware applications that address real-world requirements.

Overall, InterviewGen AI serves as an interactive learning platform that helps students and job seekers prepare for recruitment processes in a structured manner. It aims to reduce the challenges associated with independent interview preparation by making AI-assisted practice and feedback more accessible. The project establishes a foundation for further improvements in intelligent interview systems and demonstrates the potential of Generative AI in supporting students' professional development and career readiness.

### 7.2 Future Scope

The future scope of InterviewGen AI includes:

1. Advanced RAG Implementation: Improve information retrieval using vector-based search.
2. Multilingual Support: Enable interview practice in multiple languages.
3. Advanced Speech Analysis: Evaluate pronunciation, fluency, speaking speed, and filler words.
4. Adaptive Interviews: Generate questions dynamically based on candidate performance.
5. Detailed Analytics: Provide long-term performance tracking and progress comparison.
6. Expanded Job Roles: Include more industry-specific technical and non-technical interview categories.

## 8.References

1. P. Lewis, E. Perez, A. Piktus, et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020. https://arxiv.org/abs/2005.11401.

2. Y. Gao, Y. Xiong, X. Gao, et al., “Retrieval-Augmented Generation for Large Language Models: A Survey,” 2023. https://arxiv.org/abs/2312.10997.

3. V. Karpukhin, B. Oguz, S. Min, et al., “Dense Passage Retrieval for Open-Domain Question Answering,” Proceedings of EMNLP, 2020. https://arxiv.org/abs/2004.04906.

4. A. Vaswani, N. Shazeer, N. Parmar, et al., “Attention Is All You Need,” Advances in Neural Information Processing Systems (NeurIPS), 2017. https://arxiv.org/abs/1706.03762.

5. T. B. Brown, B. Mann, N. Ryder, et al., “Language Models are Few-Shot Learners,” Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020. https://arxiv.org/abs/2005.14165.

6. N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), 2019. https://arxiv.org/abs/1908.10084.